GESIS Leibniz-Institut für Sozialwissenschaften

GESIS Methods Hub

– the collaborative online portal for finding and sharing computational methods in the social sciences

Leibniz Gemeinschaft

Johannes Kiesel

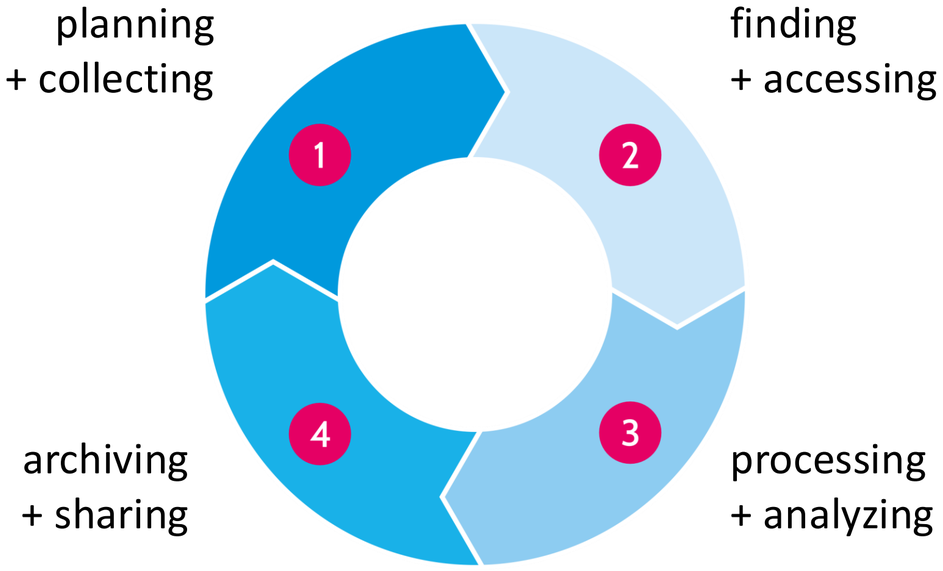
NFDI4DS Lecture Series – 2026-04-28

GESIS – Leibniz Institute for the Social Sciences

Founded in 1986, now one of the 96 Leibniz Association infrastructure institutes throughout Germany

- One of the world's leading infrastructure institutions for the social sciences
- 370+ employees: social scientists, computer scientists, and administrative staff
- We support in researching socially relevant questions with our services and own research

Research data cycle

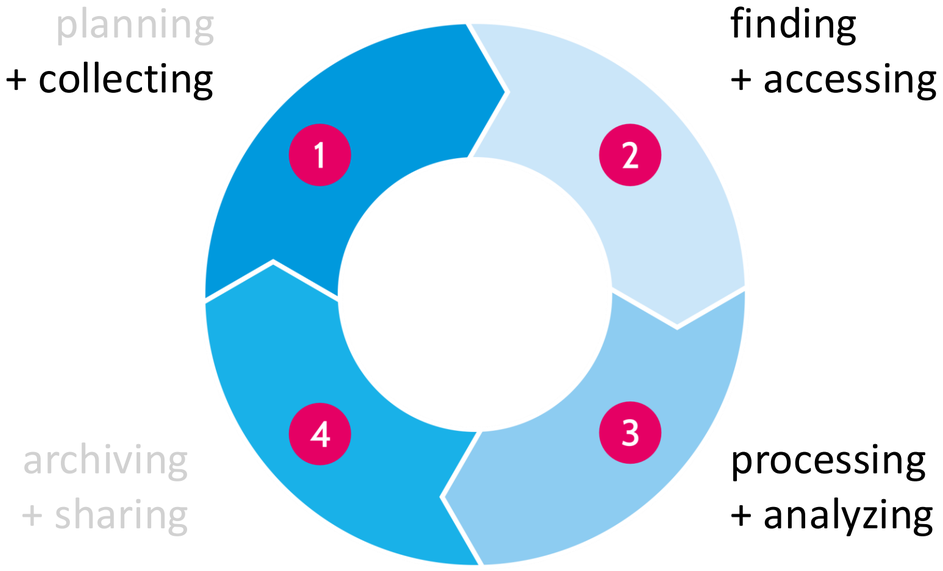


The Methods Hub

Methods Hub
@GESIS

- Open community portal of tools and tutorials to solve social science research problems
- Place to explore and apply computational approaches in interactive environments
- Features methods from data collection and preprocessing to analysis, visualization, and validation

Research data cycle



Methods Hub Core Team

Methods Hub Content

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Felix Münch

Chung-hong Chan

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Methods Hub Portal

Johannes Kiesel

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Ran Yu

Ghazal Randhawa

Raniere Costa da Silva

And support from DFG ConnOSS, Jupyter4NFDI, NFDI4DS, EU TIER2, …

Methods Hub Mission

A portal for social scientists that...

... offers high-quality content in open access format

… collaborative: from the community, to the community

... is designed to lower the technical hurdles for the use of computational analysis methods

... supports the development of reproducible workflows in data analysis

… aspires a long-term perspective as a social science informational infrastructure

Methods Hub Content

Methods Hub

Methods Hub (beta)

Methods

- computational procedure employed to address social science problems
- reusable, free, and open source software

Tutorials

- written, recorded and/or interactive introduction to a workflow or method
- building on reusable and open methods

Both often are from data science and use artificial intelligence!

Methods Hub Content

Data Collection

Pre-Processing

Data Analysis

Visualization

Evaluation & Validation

Tutorial:
Text Preprocessing To
Linzbach, Stephan
Description: Learn the basics
Topics: Text Preprocessing , P
Programming language: Pyth

Tutorial:
A Practical Guide to M (RoBERTa) Classificati
Linzbach, Stephan
Description: Learn how to cus a large language model for tex
Topics: Text Classification , Da
Programming language: Pyth

Tutorial:
Network Visualization
Schoch, David
Description: Learn how to visu network analysis
Topics: Data Visualization , Ne
Programming language: R

Tutorial:
Comparing Labeling Behaviour of Annotator Groups
Lisboa dos Santos Cotovio, Ana Maria ; Romberg, Julia ; Maurer, Maximilian
Description: Learn how to measure and interpret group-specific annotation behavior
Topics: Data Analysis
Programming language: Python

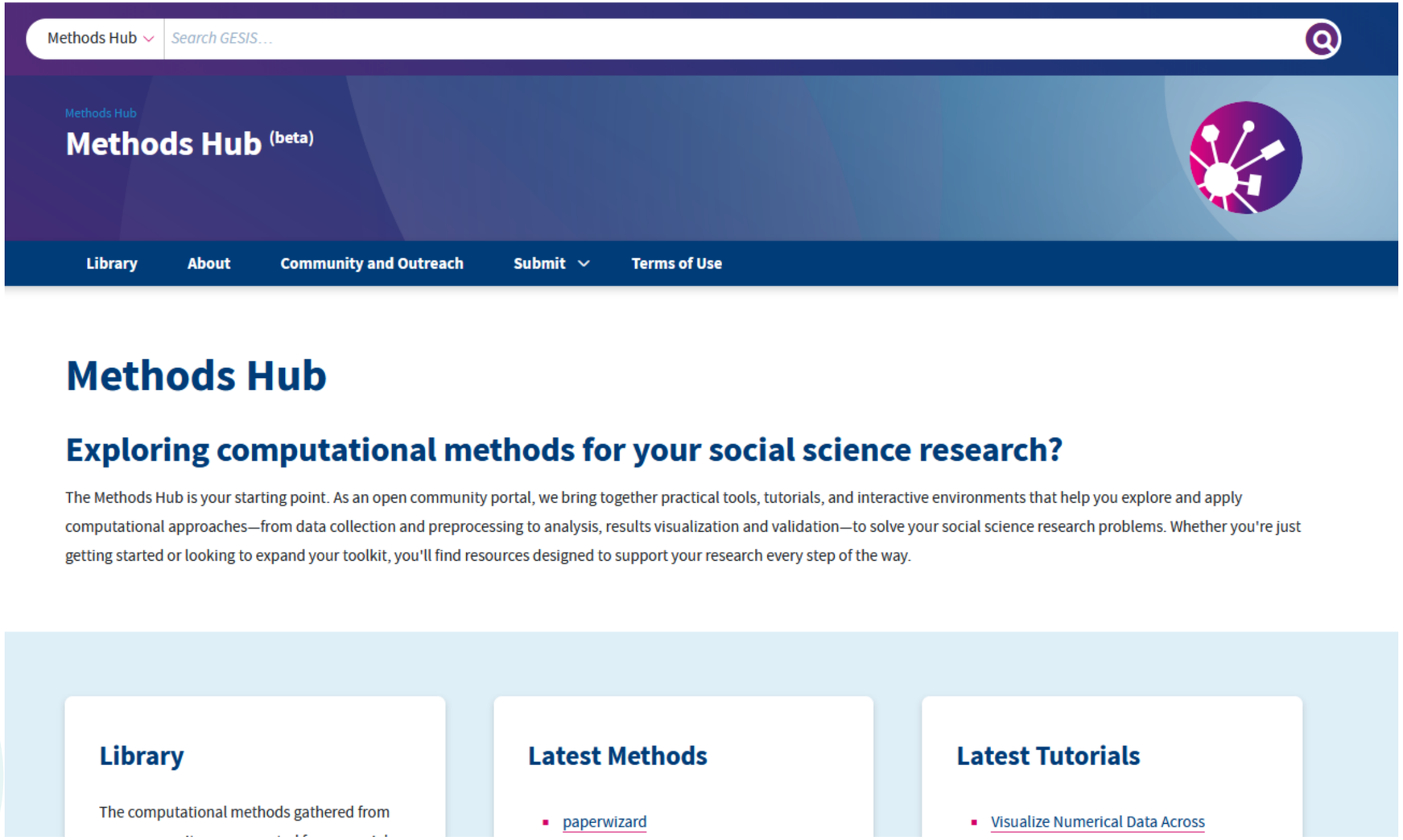
Methods Hub Components

Content
methods & tutorials

Web Portal
search, access, learn & cite

Interactive Environments
execute & test

Demo


Methods Hub
Search GESIS…
Methods Hub
Methods Hub (beta)
Library
About
Community and Outreach
Submit
Terms of Use
Methods Hub
Exploring computational methods for your social science research?
The Methods Hub is your starting point. As an open community portal, we bring together practical tools, tutorials, and interactive environments that help you explore and apply computational approaches—from data collection and preprocessing to analysis, results visualization and validation—to solve your social science research problems. Whether you're just getting started or looking to expand your toolkit, you'll find resources designed to support your research every step of the way.
Library
The computational methods gathered from
Latest Methods
paperwizard
Latest Tutorials
Visualize Numerical Data Across

Methods Hub Content Flow

Library to explore, find, and access various…

➢ACCESS

➢SEARCH

➢LEARN

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➢CITE

Content Contributor

Methods Hub Submission System

➢SHARE

Methods Hub Portal

User

Guidelines & **templates** to make submission as easy as possible while ensuring high quality and reproducibility of content

Methods Hub's Guidelines

Here you will find the guidelines used by Methods Hub.

```
# README Template: Specification and Illustration of the Methods Hub Friendly README
<!--
General specifications:
- This specification of the Methods Hub friendly README often uses the word 'should' to indi
```

➢EXECUTE

Methods Hub Interactive Environments

Methods Hub User Segments

End Users	Contributors	Multipliers

Interactive Execution Backends

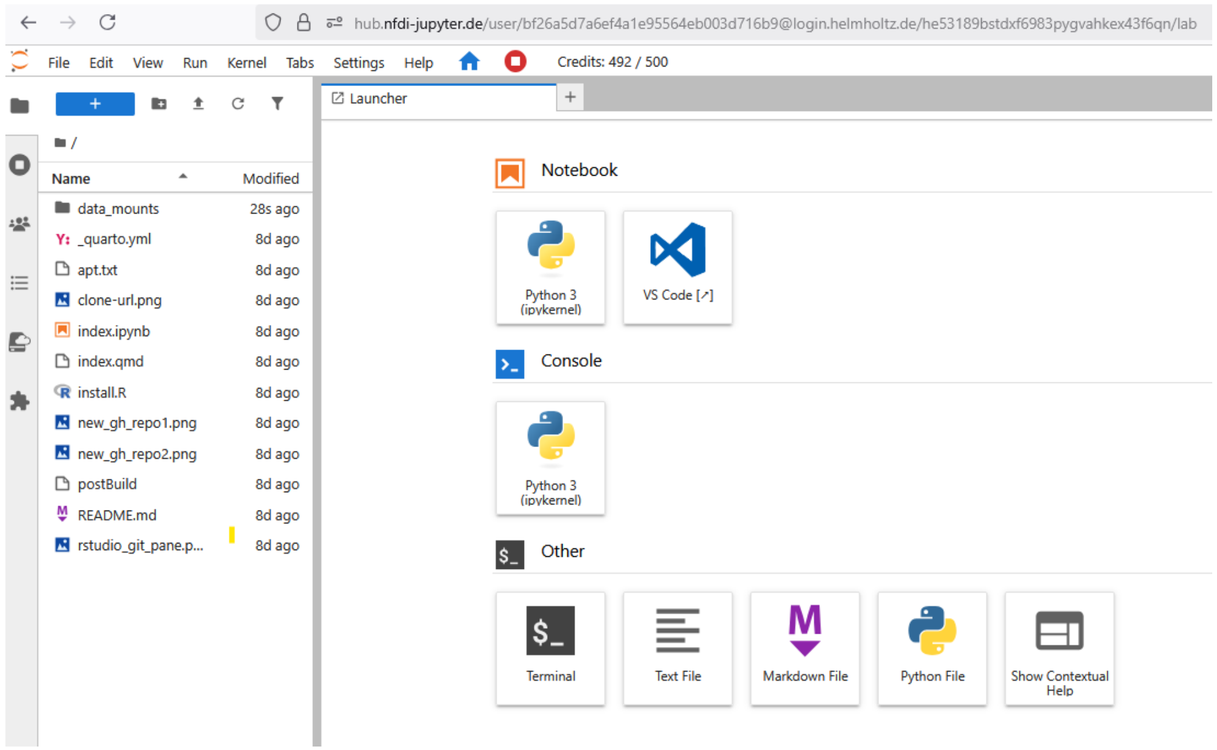
binder

Open in an Interactive Environment

- Open in MyBinder
- Open in Jupyter4NFDI

Jupyter®4nfdi

This work is supported by Jupyter4NFDI as part of Base4NFDI (DFG project no.521453681).



User 1: Try it out
User 2: Conduct full analysis
User 3: Establish reproducibility

Methods Hub Outlook

We are currently working on:

- Fully open submission of content
- Automated quality checks of content before and after submission
- Easy submission of prompts as methods
- Streamlining use of interactive environments
- Empirically based task taxonomy
- Benchmarking, method writing support, …

Meet us here:

- 2026-05 21+22. Method Hub Updates @ NFDI4DS meeting in Cologne.
- 2026-05-26. Methods Sharing Retreat @ WebSci‘26 in Braunschweig.
- 2026-06 3+4. Methods Hub Started Bundles @ ICA‘26 Hackathon in Cape Town.
- 2026 06-29 – 07-03. R2CASS Workshop @ WebConf’26 in Dubai or online.

and more content! what about yours? write us at

methodshub@gesis.org

and 24/7 at

https://methodshub.gesis.org